J08: CENTRALITY OF THE FULL TWIST AND CONJUGACY OF COXETER WORDS IN THE POSITIVE BRAID MONOID

Abstract. We give the rewriting sequences behind job J08 (record M408) of the dependency audit. On m strands, with $c = \sigma_0\sigma_1\cdots\sigma_{m-2}$ and $\Delta^2 = c^m$, we show that $\Delta^2\sigma_i$ and $\sigma_i\Delta^2$ are related by far commutations and positive braid relations applied inside words, for every generator σ_i. We then show that for $m \geq 2$ every word γ containing each generator exactly once satisfies $\Delta^2\gamma \sim \Delta^2 c$, where $\sim$ is the relation `PositiveWordConj`. For $m \leq 1$ the second statement is false, because the only word is the empty word and `PositiveWordConj` relates no word to the empty word. All proofs are elementary and self-contained.

1. Words and relations

Fix an integer $m \geq 0$ and put $n := \max(m-1, 0)$, the value of the truncated difference $m-1$ in the Lean code. The *alphabet* is $\{0, 1, \ldots, n-1\}$, and the letter i stands for the positive generator σ_i of the braid group on m strands. A *word* is a finite sequence of letters (the Lean type `PositiveAdjacentWordm`, a list of elements of `Fin(m-1)`). Juxtaposition uv denotes concatenation, $\varnothing$ is the empty word and $\mathrm{len}(u)$ is the length of u. Two letters i, j are *far* if $|i-j| \geq 2$. For a letter x and a word w we write $x \perp w$ if x is far from every letter of w.

Definition 1.1. An *elementary braid move* replaces a word $u\,i\,j\,v$ with i, j far by $u\,j\,i\,v$, or a word $u\,i\,(i+1)\,i\,v$ by $u\,(i+1)\,i\,(i+1)\,v$, where u, v are arbitrary words and $i+1 \leq n-1$ in the second case. A *rotation* replaces a word $u\,i$ by $i\,u$, where i is a letter.

We use three relations on words.

(1) *Positive braid equivalence* $u \equiv v$ is the equivalence relation generated by elementary braid moves (Lean: `PositiveBraidEquiv`, the closure `Relation.EqvGen` of `PositiveBraidStep`).
(2) *Positive word conjugacy* $u \sim v$ is the relation `PositiveWordConj` of `PositiveWordNormalForms.lean`: the smallest relation containing all pairs $(u\,i,\ i\,u)$ given by rotations and all pairs (u', u'') given by elementary braid moves, and closed under symmetry and transitivity. Reflexivity is not part of the definition.
(3) *Far rotation equivalence* $u \approx v$ is the equivalence relation generated by elementary braid moves of the first kind (far commutations) and by rotations (Lean: `FarRotationEquiv`).

Lemma 1.2. *Let u, v be words.*

(1) *If $u \equiv v$, then $a\,u\,b \equiv a\,v\,b$ for all words a, b.*
(2) *If $u \sim v$, then $\mathrm{len}(u) = \mathrm{len}(v) \geq 1$. In particular $\varnothing \sim v$ holds for no word v.*
(3) *If $u \neq \varnothing$, then $u \sim u$.*
(4) *If $u \equiv v$ and $u \neq \varnothing$, then $u \sim v$.*

Proof. (1) An elementary braid move from u to v remains an elementary braid move from aub to avb, after the outer words of Definition 1.1 are enlarged by a and b. The set of pairs (u, v) with $aub \equiv avb$ for all a, b is therefore an equivalence relation containing the elementary braid moves, and it contains $\equiv$.

Date: 2 November 2026.

(2) Rotations and elementary braid moves preserve the length, and each of them relates two nonempty words. Symmetry and transitivity preserve both properties.

(3) Write $u = u'x$ with a letter x. The rotation gives $u \sim xu'$, symmetry gives $xu' \sim u$, and transitivity gives $u \sim u$.

(4) The set of pairs (u, v) with $\mathrm{len}(u) = \mathrm{len}(v)$ and with $u \sim v$ whenever $u \neq \varnothing$ is an equivalence relation by (3) and contains the elementary braid moves. Hence it contains $\equiv$. □

Remark 1.3. Rotations enter the proof of Corollary 2.4 below only through Lemma 1.2(3), which provides $u \sim u$ for the reflexivity steps of $\equiv$. Every other step is an elementary braid move inside a word. If $m = 2$, the alphabet is $\{0\}$, no elementary braid move exists, $\equiv$ is equality, and the relation $u \sim u$ cannot be obtained without a rotation.

Lemma 1.4. *Let x be a letter and w a word with $x \perp w$. Then $w\,x \equiv x\,w$ and $w\,x \approx x\,w$, and the same equivalences hold after adding arbitrary words on the left and on the right.*

Proof. Induction on $\mathrm{len}(w)$. For $w = y\,w'$ the induction hypothesis gives $y\,w'\,x \equiv y\,x\,w'$, and the far commutation of y and x gives $y\,x\,w' \equiv x\,y\,w'$. The resulting chain from $w\,x$ to $x\,w$ consists of far commutations. A far commutation remains a far commutation after arbitrary words are added on the left and on the right, so the chain also proves $a\,w\,x\,b \equiv a\,x\,w\,b$ and $a\,w\,x\,b \approx a\,x\,w\,b$ for all words a, b. □

2. Centrality of the full twist

Assume $n \geq 1$, that is, $m \geq 2$. Let $c := 0\,1\,\cdots\,(n-1)$ be the Coxeter word (`coxeterWordm`), let c^k be its k-th power (`torusWordmk`), and let $\Delta^2 := c^m = c^{n+1}$ (`fullTwistWordm`). For a word w whose letters are at most $n-2$, let w^+ be the word obtained by adding 1 to every letter. Put

$$\hat{c} := 0\,1\cdots(n-2), \qquad \tilde{c} := 1\,2\cdots(n-1),$$

so that $c = \hat{c}\,(n-1) = 0\,\tilde{c}$ and $\hat{c}^{\,+} = \tilde{c}$. For $n = 1$ both words are empty.

Lemma 2.1. (1) *For every letter $i \leq n-2$, $c\,i \equiv (i+1)\,c$.*

(2) *For every word w whose letters are at most $n-2$, $c\,w \equiv w^+\,c$.*

(3) *For all $k \geq 0$ and every letter i with $i + k \leq n-1$, $c^k\,i \equiv (i+k)\,c^k$.*

Proof. (1) Write $c = A\,i\,(i+1)\,B$ with $A = 0\cdots(i-1)$ and $B = (i+2)\cdots(n-1)$. Every letter of B is at least $i+2$, so $i \perp B$, and every letter of A is at most $i-1$, so $(i+1) \perp A$. By Lemma 1.4, the braid move, and Lemma 1.4 again,

$$c\,i = A\,i\,(i+1)\,B\,i \;\equiv\; A\,i\,(i+1)\,i\,B \;\equiv\; A\,(i+1)\,i\,(i+1)\,B \;\equiv\; (i+1)\,A\,i\,(i+1)\,B = (i+1)\,c.$$

(2) Induction on $\mathrm{len}(w)$. For $w = x\,w'$, part (1) and the induction hypothesis give $c\,x\,w' \equiv (x+1)\,c\,w' \equiv (x+1)\,w'^{+}\,c = w^+c$, by Lemma 1.2(1).

(3) Induction on k, for all i at once. The case $k = 0$ is trivial. If $i+k+1 \leq n-1$, then $i \leq n-2$, and part (1) and the induction hypothesis applied to the letter $i+1$ give

$$c^{k+1}\,i = c^k\,c\,i \;\equiv\; c^k\,(i+1)\,c \;\equiv\; (i+1+k)\,c^k\,c = (i+k+1)\,c^{k+1}.$$ □

Lemma 2.2. $c^2\,(n-1) \equiv 0\,c^2$.

Proof. The letters of $\hat{c}$ are at most $n-2$, so Lemma 2.1(2) gives $c\,\hat{c} \equiv \tilde{c}\,c$. Hence, by Lemma 1.2(1),

$$0\,c^2 = 0\,c\,\hat{c}\,(n-1) \;\equiv\; 0\,\tilde{c}\,c\,(n-1) = c\,c\,(n-1) = c^2\,(n-1).$$ □

Theorem 2.3. *For every letter i, $\Delta^2\,i \equiv i\,\Delta^2$. Consequently $u\,x\,\Delta^2\,v \equiv u\,\Delta^2\,x\,v$ for all words u, x, v.*

Proof. Let $k := n-1-i$, so that $\Delta^2 = c^{n+1} = c^i\,c^2\,c^k$. Lemma 2.1(3) with the letter i and exponent k, Lemma 2.2, and Lemma 2.1(3) with the letter 0 and exponent i give

$$\Delta^2\,i = c^i\,c^2\,c^k\,i \;\equiv\; c^i\,c^2\,(n-1)\,c^k \;\equiv\; c^i\,0\,c^2\,c^k \;\equiv\; i\,c^i\,c^2\,c^k = i\,\Delta^2,$$

where each step uses Lemma 1.2(1). The second statement follows by induction on $\mathrm{len}(x)$: for $x = y\,x'$, $u\,y\,x'\,\Delta^2\,v \equiv u\,y\,\Delta^2\,x'\,v \equiv u\,\Delta^2\,y\,x'\,v$. □

Corollary 2.4. *Let $m \geq 0$. For all words u, v and every letter i, $u\,i\,\Delta^2\,v \sim u\,\Delta^2\,i\,v$. If $m \geq 2$, then $u\,x\,\Delta^2\,v \sim u\,\Delta^2\,x\,v$ for all words u, x, v.*

Proof. If a letter exists, then $m \geq 2$, and the first word is nonempty. Theorem 2.3 and Lemma 1.2(4) give the first statement. If $m \geq 2$, then $\mathrm{len}(\Delta^2) = m(m-1) \geq 2$, and the second statement follows in the same way. If $m \leq 1$, no letter exists and the first statement is vacuous. □

3. Conjugacy of Coxeter words

A *Coxeter word* is a word containing every letter $0, \ldots, n-1$ exactly once, that is, a permutation of the list c. For $0 \leq k \leq n$ let $s_k := 0\,1\cdots(k-1)$, the first k letters of c (Lean: `(List.finRange (m-1)).takek`), so that $s_n = c$ and $s_{k+1} = s_k\,k$.

Remark 3.1. The audit cites Humphreys, *Reflection Groups and Coxeter Groups*, §3.16, for the conjugacy of Coxeter elements in a Coxeter group whose Coxeter graph is a tree. That reference was not consulted for this note. Proposition 3.4 below is proved directly for the path $0-1-\cdots-(n-1)$, by induction on the number of letters instead of by the source and sink argument.

Lemma 3.2. (1) *If $u \approx v$, then v is a permutation of u.*
(2) *For all words α, β, $\alpha\,\beta \approx \beta\,\alpha$.*

Proof. (1) Far commutations and rotations permute letters. (2) Induction on $\mathrm{len}(\beta)$, for all α at once. For $\beta = b\,\beta'$, the induction hypothesis applied to αb and the inverse of a rotation give $\alpha\,b\,\beta' \approx \beta'\,\alpha\,b \approx b\,\beta'\,\alpha$. □

Lemma 3.3. *Let $t \leq n-1$ be a letter and let δ, δ' be words with $\delta \approx \delta'$. Assume that the letters of δ are pairwise distinct and smaller than t. Then $\delta\,t \approx \delta'\,t$.*

Proof. By Lemma 3.2(1), every word in a chain of moves from δ to δ' has the two properties assumed for δ. It is therefore enough to treat the case in which δ' is obtained from δ by one far commutation or one rotation. A step of the chain in the opposite direction is handled by this case applied to the reversed step, whose initial word has the same two properties, followed by symmetry of $\approx$. A far commutation $\delta = u\,i\,j\,v \to u\,j\,i\,v = \delta'$ gives the far commutation $u\,i\,j\,v\,t \to u\,j\,i\,v\,t$. Let $\delta = u\,i$ and $\delta' = i\,u$. If $i \leq t-2$, then i and t are far and

$$u\,i\,t \;\approx\; u\,t\,i \;\approx\; i\,u\,t.$$

Otherwise $i = t-1$. The letters of u are smaller than t and different from i, so they are at most $t-2$ and $t \perp u$. Two rotations and Lemma 1.4 give

$$u\,i\,t \;\approx\; t\,u\,i \;\approx\; i\,t\,u \;\approx\; i\,u\,t. \qquad \square$$

Proposition 3.4. *For $0 \leq k \leq n$, every permutation γ of s_k satisfies $\gamma \approx s_k$. In particular every Coxeter word γ satisfies $\gamma \approx c$.*

Proof. Induction on k. For $k = 0$, $\gamma = \varnothing = s_0$. Let $k+1 \leq n$ and let γ be a permutation of $s_{k+1} = s_k\,k$. Write $\gamma = \alpha\,k\,\beta$. Lemma 3.2(2) applied to αk and β gives $\gamma \approx \beta\,\alpha\,k$. By Lemma 3.2(1), the word $\delta := \beta\,\alpha$ is a permutation of s_k, and the induction hypothesis gives $\delta \approx s_k$. The letters of δ are pairwise distinct and smaller than k, so Lemma 3.3 with $t = k$ gives $\delta\,k \approx s_k\,k = s_{k+1}$. □

Theorem 3.5. *Let $m \geq 2$ and let γ be a Coxeter word. Then $\Delta^2\gamma \sim \Delta^2 c$. If $m \leq 1$, then $\Delta^2\gamma \sim \Delta^2 c$ is false.*

Proof. Let $m \geq 2$. We show that $\gamma \approx \gamma'$ implies $\Delta^2\gamma \sim \Delta^2\gamma'$. The set of pairs (γ, γ') with $\Delta^2\gamma \sim \Delta^2\gamma'$ is symmetric and transitive, and it is reflexive by Lemma 1.2(3), because $\Delta^2 \neq \varnothing$. A far commutation $\gamma = u\,i\,j\,v \to u\,j\,i\,v$ gives the far commutation $\Delta^2 u\,i\,j\,v \to \Delta^2 u\,j\,i\,v$. For a rotation $u\,i \to i\,u$, a rotation and Corollary 2.4 give

$$\Delta^2\,u\,i \;\sim\; i\,\Delta^2\,u \;\sim\; \Delta^2\,i\,u.$$

Proposition 3.4 gives $\gamma \approx c$, hence $\Delta^2\gamma \sim \Delta^2 c$.

If $m \leq 1$, the alphabet is empty, so $\gamma = c = \Delta^2 = \varnothing$, and Lemma 1.2(2) shows that $\varnothing \sim \varnothing$ is false. □

4. Invariants of positive word conjugacy

For a word $u = x_1 \cdots x_r$ let $\pi(u) := \tau_{x_1} \circ \cdots \circ \tau_{x_r}$, where τ_i is the transposition of the strands i and $i+1$ (Lean: `PositiveAdjacentWord.permutation`). Let $L(u)$ be the set of letters of u (Lean: `u.toSigned.generatorIndices`), and let $\kappa(u)$ be the number of orbits of the cyclic group generated by $\pi(u)$ on $\{0, \ldots, m-1\}$, fixed points included (Lean: `u.toSigned.cycleCount`).

Proposition 4.1. *If $u \sim v$, then $\mathrm{len}(u) = \mathrm{len}(v)$, $L(u) = L(v)$, the permutations $\pi(u)$ and $\pi(v)$ are conjugate, and $\kappa(u) = \kappa(v)$. If $u \equiv v$, then $\pi(u) = \pi(v)$.*

Proof. It suffices to check one rotation and one elementary braid move, since all four properties define equivalence relations. A rotation permutes letters, and $\pi(i\,u) = \tau_i\,\pi(u) = \tau_i\,(\pi(u)\,\tau_i)\,\tau_i^{-1} = \tau_i\,\pi(u\,i)\,\tau_i^{-1}$. A far commutation permutes letters, and $\tau_i\tau_j = \tau_j\tau_i$ for far i, j because the two transpositions have disjoint supports. A braid move replaces $i\,(i+1)\,i$ by $(i+1)\,i\,(i+1)$, which has the same set of letters, and both products $\tau_i\tau_{i+1}\tau_i$ and $\tau_{i+1}\tau_i\tau_{i+1}$ equal the transposition of i and $i+2$. Finally, if $\pi' = g\pi g^{-1}$, then g maps each orbit of π onto an orbit of π', so κ is a conjugacy invariant. □

5. Correspondence with the Lean declarations

The modules `OnLeConjecture.KnotTheory.PositiveWordBraidEquiv` and `OnLeConjecture.KnotTheory.PositiveWordCoxeterConjugacy` formalize the statements above.

Statement	Lean declaration
Lemma 1.2(1)	`PositiveBraidEquiv.append_left`, `PositiveBraidEquiv.append_right`, `PositiveBraidEquiv.append`, `PositiveBraidEquiv.append_middle`
Lemma 1.2(2)	`PositiveWordConj.length_eq`, `PositiveWordConj.ne_nil_left`, `not_positiveWordConj_nil_left`
Lemma 1.2(3)	`PositiveWordConj.refl_of_ne_nil`
Lemma 1.2(4)	`PositiveBraidEquiv.positiveWordConj`, `PositiveBraidEquiv.positiveWordConj_append_middle`
Lemma 1.4	`PositiveBraidEquiv.append_singleton_of_far`, `FarRotationEquiv.pass_far`
Lemma 2.1	`coxeterWord_append_singleton`, `coxeterWord_append_shift`, `torusWord_append_singleton`
Lemma 2.2	`coxeterWord_wrap`

Theorem 2.3	`torusWord_strandCount_append_singleton`, `fullTwistWord_append_singleton`, `fullTwist_central_braidEquiv`, `fullTwist_commute_braidEquiv`
Corollary 2.4	`fullTwist_central`, `fullTwist_commute`
Lemma 3.2	`FarRotationEquiv.perm`, `FarRotationEquiv.append_comm`
Lemma 3.3	`FarRotationStep.append_singleton_lift`, `FarRotationEquiv.append_singleton_lift`
Proposition 3.4	`farRotationEquiv_take_finRange`, `farRotationEquiv_coxeterWord`
Theorem 3.5	`FarRotationEquiv.fullTwist_positiveWordConj`, `fullTwist_coxeter_conj`, `fullTwist_coxeter_conj_of_perm`, `not_positiveWordConj_of_le_one`, `fullTwist_coxeter_conj_iff`
Proposition 4.1	`PositiveWordConj.length_eq`, `PositiveWordConj.generatorIndices_eq`, `PositiveWordConj.isConj_permutation`, `PositiveWordConj.cycleCount_eq`, `PositiveBraidEquiv.permutation_eq`, `SignedAdjacentWord.cycleCount_eq_of_isConj`

Remark 5.1. The audit proposes `fullTwist_coxeter_conj` without a hypothesis on m. By Theorem 3.5 that statement is false for $m \in \{0, 1\}$. The Lean theorem `fullTwist_coxeter_conj` therefore carries the hypothesis $2 \leq m$ as its last argument, and `fullTwist_coxeter_conj_iff` records that the conclusion holds exactly when $2 \leq m$. The proposed `fullTwist_central` is proved without change.